

Ujjwal Agrawal

ujjwalagrawal3012@gmail.com | +91-8763343874 | [Portfolio](#) | [LinkedIn](#)

EDUCATION

Ramaiah Institute of Technology | Aug '18 – Jul '22 | Bachelor of Engineering | 8.6/10

EXPERIENCE

Software Engineer 2 – Simply Vyapar Apps | July '24 – Present

- **Reduced database queries by 70% and improved page load by 3x (2.1s → 0.7s)** by architecting a sync-aware, DB-transaction-aligned hybrid cache invalidation system that replaced inefficient full-reload patterns with selective refresh logic.
- **Eliminated 100% of partial commits and reduced transaction-related defects by 85%** by designing a distributed transaction framework enforcing atomic commits between local and server databases (commit-on-remote-success model).
- **Reduced PDF-related support tickets by 75% and eliminated non-recoverable production failures** by designing a resilient PDF generation pipeline with multi-engine fallback (native print API → headless browser).
- **Reduced HTML payload size by 66% and improved preview stability by 50%** by redesigning the invoice/PDF generation framework in an Electron desktop app, replacing DOM-bound rendering with IPC-based file-backed previews.
- **Standardized UI architecture across 10+ business modules** by architecting a reusable composite page-layout framework (React + TypeScript) with modular, typed sections (titular, stats, filters, table, detail, common states), cutting per-page development effort by 25% and eliminating duplicate frontend code via plug-and-play integration.
- **Eliminated full table scans on large datasets** by building a two-phase Master Search a pre-built FTS3 index resolves matches across 9+ fields in $O(\log n)$ time, and only matched IDs are passed to a targeted paginated query with date, type, and role filters.
- **Prevented UI crashes on large datasets** by engineering a bidirectional infinite scroll model that keeps only 70 rows in memory at any time, evicting old blocks on forward scroll and re-fetching on backward scroll with reactive datasource resets ensuring data stays fresh on every filter, sort, or search change, and full parity maintained across Excel and print exports.
- **Improved dashboard responsiveness by 40% and reduced unnecessary re-renders by 30%** by architecting a real-time dashboard framework with centralized state management, enabling dynamic widget selection and multi-range sales analytics.
- **Improved user activation by 30% and reduced onboarding drop-off by 15%** by building a Firebase Remote Config-driven A/B experimentation system that split first-time users between a demo-scheduling flow and a guided first-sale walkthrough, with phone/OTP authentication and a skippable gate controlling entry.
- **Built role-based item visibility controls for salesman management** admins can restrict salesmen to specific item categories, with smart defaulting logic ensuring newly added categories automatically inherit the correct access without manual reassignment, backed by DB schema changes and a redesigned admin UI.

Software Engineer 1 – BizAcuity Solutions | July '22 – July '24

- **Improved development velocity by 20%** by leading the frontend team in migrating the codebase to React.js and TypeScript, standardizing component architecture across the application.
- **Drove 10% business growth** by building a bulk user onboarding feature enabling CSV-based imports, significantly reducing manual setup time and improving customer adoption.
- **Increased user conversion by 30%** by redesigning the website signup and login flow, implementing secure authentication, optimized validation, and friction-reducing UX improvements.
- **Reduced incident response time by 30%** by building an event-driven alerting system with automated email notifications and centralized alert tracking.
- **Reduced page load time by 30%** by translating UI/UX designs into optimized frontend implementations and applying performance optimization strategies to improve user experience.
- Streamlined client onboarding by designing database schemas, building a Python-based Kafka consumer to process event streams, and integrating client data into the platform using optimized SQL workflows.

ACHIEVEMENT

- Won **Simply Vyapar Tech Hackathon 2025** (Generative AI track), competed against internal engineering teams, building an AI-powered solution judged on technical depth and business impact.

SKILLS

Frontend: React.js, TypeScript, JavaScript ES6+, HTML5, CSS3, Redux/Redux Toolkit, Formik, AG Grid, Firebase, Material UI, Bootstrap, SASS, Radix, Electron

Backend: Java, Spring Boot, REST APIs

Databases: SQL, MySQL, Vertica, SQLite

Architecture: Distributed Systems, Transaction Management, Caching, State Management, ACID

Analytics & Tools: Mixpanel, Clevertap, GA, Sentry, Git, Figma, Jira